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PLANTATION JOURNAL



Sustainable Forests & Livelihoods (SUFAL) Project

“Jhaw Plantation”

Range : Nalchira Range, Hatiya.

Beat : Dhal Char Beat.

Plantation Area : 50.0 Hectare.

Financial Year : 2022-2023

Coastal Forest Division, Noolkholi

Raising Jhaw of 50.0 Hectare. Plantation in Financial Year 2022-2023 under SUFAL Project.

A Basic Information			
1.	Name of the plantation with Year	Jhaw Plantation, FY : 2022-2023	
2.	Plantation Area	50.0 Hectare	
3.	Beat, Range & Divison	Dhal Char Beat, Nalchira Range, Coastal Forest Division Noakhali	
4.	Species	<u>Casuarina equisetifolia</u> (Jhaw),	
5.	No. of seedlings	1,25,000 Nos.	
6.	Location	District: Noakhali	Union: Sukh Char
		Upazila: Hatiya	Char: Char ^a Ayesa
7	GPS Reading	8 A. N 22°21'06.2 B. N 22°20'45.8 C. N 22°20'34.3 D. N 22°20'58.4 E 91°03'36.3 E 91°03'37.2 E 91°03'59.7 E 91°04'06.9	
8.	Name & Designation of the Officers & stuffs Involved in the Plantation Raising Activitiest	1. Mr. Md. Farid Meah, DCF. 2. Mr. Kazi Tariqur Rahman, ACF. 3. Md. Alauddin, Forester, Range Officer 4. Md Faruk Hossain, Forester, Beat Officer. 5. Mr. Samim Al Mamun, Forest Guard. 6. Mr. Babul hoque, Forest Guard. 7. Mr. Jamshed uddin Seddek, Plantation Male.	
B Characteristic of the Plantation Site			
1.	Area of the Plantation	50.0 ha Jhaw Plantation in the year of 2022-2023.	
2	Edaphic Condition	Physiographically the Char is part of the young Meghna Estuarine Flood Plain. Hobgoblin, the site is soft and clayey and it is also comparatively compact and harder as well as at the same time subjected to more saline condition. These soils permit easy penetration of mound and mangrove tree roots because of being quite fertile.	
3	Climate Condition	This char has a tropical monsoon climate, characterized by basically four seasons per year. The average temperature usually ranges between 11 ° to 29 ° C in winter and 21 ° to 38 ° C during summer. The average	

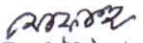
Selection Criteria of the Planting Site

The plantation site has been selected through the following criteria.

- a. Land suitability: The land of the planting site (Jahajmara Sadar) is hard and Sandy Loam. In addition, the site is comparatively compact and harder as well as at the same time subject to more saline condition where Jhaw seedlings e.g. Casuarina equisetifolia (Jhaw) are suitable for plantation.
- b. Grazing Problem: It is known to that there entire are a number of chars in Noakhali and accreting new lands. These chars are threat to the grazing problems. But this is free from such unavoidable phenomenon. That is why this site is selected for the mangrove plantation.
- c. Biodiversity Conservation: The Coastal afforestation program will bring a result in the establishment of plantation rich in biodiversity and higher density of wildlife habitats. Natural regeneration of forest tree species with their associates will be enhanced. In addition, through the project area the condition of coastal forest in Bangladesh will extent.
- d. Poverty Reduction: The project activities are comprised of civil works, nursery and plantation raising and maintenance, gardening etc.; which are labor intensive and technical works. Male and female can be deployed or engaged in all these works based on their working capability; which will create employment opportunities for them and help them in alleviating poverty and fulfilling needs. Thus they will be benefited socio-economically and their social esteem will be elevated.
- e. Sectoral Target: The people residing in the project area are exposed to frequent cyclones and tidal surge which makes them highly vulnerable to recent climate change effects. As a result, their socio-economic condition is less secured. Beside this, the land mass developed due to accretion of the sediments discharged through Meghna estuary needs to be stabilized. On the other hand, the objectives of the forestry sector is to increase the total forest cover of the country and to optimize sustainable production of goods and services from the existing limited forest resources. The coastal area of the proposed project is such type of land where forestry resources could be developed through establishment of plantations.
- f. National Strategy: The proposed project is also aligned with the National Strategy; the *Bangladesh Sixth Five Year Plan (FY2019-FY2023)*. According to the plan, existing scattered coastal areas will be planted to increase productivity by promoting scientific management principles. In order to prevent the extent of damage by cyclones and tidal surges, the plan envisages creating a Coastal Green Belt and enhancing access to input materials, including seeds and seedlings. In addition, past climate disasters have shown that resilience to cyclones and storm surges is greater where mangrove greenbelts and buffer ecosystems exist. It is evident that mangroves can mitigate or reduce risk of natural disasters such as cyclones and tsunamis.
- g. Global Aspect: According to the IPCC report of 2013, approximately 95% human beings are the responsible for the climate change and Bangladesh is the home to the world's largest delta systems. Two third of the country is less than five meters above the sea level. It is vulnerable to short term climate shocks such as cyclone (DFID-2011). As a result, human beings and natural resources are threats to climate

5	Climate Resilient Aspect of the Plantation	★ The Coastal forest, breakwater, seawall is to absorb the impact forces and to retard the flow of large storm waves and cyclone. A seawall, if tall enough, reflects the waves back out to sea. On the other hand, permeable structures, like breakwaters and coastal forests, partly reflect and partly transmit the water. In the case of a coastal forest, energy is progressively absorbed as it passes through the forest. Without the forest barrier, the tsunami/cyclone will run-up to a maximum height determined by the magnitude and nature of the seismic event that created the tsunami and local factors such as the coastal profile, offshore bathymetry and beach slope that modify the wave's force. Once the tsunami comes on shore, the amount of reduction in water depth, velocity, and force depends on how much water is reflected and energy is absorbed by the coastal forest. ★ Through the afforestation in the project area biodiversity conservation, human security and so on will be ensured in the project area. Therefore, greenbelt of the project area by rising said plantation has been planned to bring back the balance of eco-system into the same char lands and to develop a friendly environment by raising coastal afforestation. It is for the fact that the land is suitable for mangrove and non-mangrove plantations where bio-physical and socio-economic conditions will improve. The Jahajmara Sadar will be brought to work as a barrier during natural disasters by dint of the mangrove plantation.
C Planting Activities		
1	Delineation of the Planting Area	The Jahajmara Sadar is the ultimate result of sedimentation lead to form a marshy land whose soils permit easy penetration of Jhau tree roots and agricultural crop roots because of being quite fertile. The proposed project will be used for different types of plantations which are suitable in the coastal zone by leveling with the existing species diversity. If the coastal forest lands of a target area are properly replenished with greens, the existing wildlife habitats will increase and will be a nice environment for Biodiversity conservation. This type of coastal forest lands can fight back the prevailing environmental threats and mitigate the negative impacts of the climate changes took place in these coastal zones. Thus, this type of coastal forest lands not only save the life and resources from natural disasters but also play important role in economic development.
2	GPS Reading of all Corners of the Planting Site	The GPS has been experiment in reaching the every position of the plantation site particularly through the each corner of the site in presence of the RO, BO, FG, PM as well as all other FD-Noakhali under Hatiya Upazila.
3	Post-Planting Survey, Surveyor, Survey Technique & Record	Mr. Md. Shafiquil Islam Surveyor The Survey has been completed through the technique of GPS reading and it is 1st recorded in the note book after that the record is shown in the Map
4	Post-Planting Map	

5. Signature of the officers Involved in the Raising Mangrove Plantation activities :

Sl.	Name	Designation	Signature
01	Mr. Md Faruk Hossain .Forester	Beat Officer, Dhal Char Beat Nalchira Range, Hatiya. Coastal Forest Division, Noakhali.	 Md Faruk Hossain Forester Beat Officer, Dhal Char Beat Nalchira Range, Hatiya. Coastal Forest Division, Noakhali..
02	Mr. Md. Alauddin. Forester	Range Officer Nalchira Range, Hatiya Coastal Forest Division Noakhali.	 Md. Alauddin Range Officer Nalchira Range, Hatiya Coastal Forest Division, Noakhali.
03	Mr. Kazi Tariqur Rahman, ACF	Assistant Conservator of Forest Coastal Forest Division Noakhali.	 Kazi Tariqur Rahman Assistant Conservator of Forest Coastal Forest Division Noakhali.
04	Mr. Md. Farid Meah, DCF.	Divisional Forest Officer Coastal Forest Division Noakhali.	 Md. Farid Meah Divisional Forest Officer Coastal Forest Division Noakhali.

EXPENDITURE SHEET

Jhaw Plantation'2022-23, Area : 50.0 Hectare.

Dhal Char Beat, Nalchira Range, Coastal Forest Division, Noakhali.

Project : Sustainable Forests & Livelihoods (SUFAL) Project

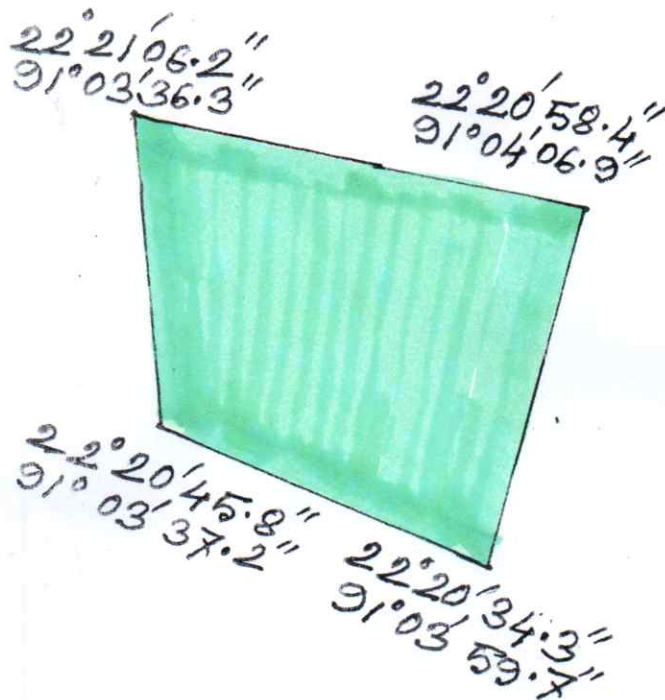
Vr. No.	Month	Particulars	Amount		NBP
			Current	Progress	
10/NC	June/2023	Dalcher Beat of Nalchira Range.for Survey the Plantation site,prepare plantatiom site map post planting survey and making of map GPS readings at every turn along the boundary of the plantation site need to be noted on the plantation journal.Stacking (130,000 stacks to be collected or purchased for sound stacking at 125,000 points at 2 m X 2m)inculding fastening of seedling with stack by shootly.pit (1' X 1' X 1')making leave the pits open for about 2 weeks 125,000 pits are required per hectarc. .procure compost or decomposed cowdung pit.mix these manures properly with the soil in the pit. pulverize the soil.max it with compost/cowdung remove all debris root pieces,plant the seedlings by taking off the bags carefully two month,Application of fertilizers after planting,post planting Weeding,post planting survey and making of map,Fixing of a RCC plantation Board (Signboard size 8'- 0" X 3'- 0"From ground 2'- 6" hight at 3'- 0" X 3'-0" for 5.0ha 1 no of signboard with writing and painting of plantation information Raising of 50.0 Hectare. Jhaw plantation in 2022-2023 financial year Under SUFAL Project Daily labours 550 M/Days @ 500/-	2,75,000/-	2,75,000/-	53/592
11/NC	June/2023	Do Daily labours 550 M/Days @ 500/-	2,75,000/-	550,000/-	53/592
12/NC	June/2023	Do Daily labours 550 M/Days @ 500/-	2,75,000/-	825,000/-	53/592
13/NC	June/2023	Do Daily labours 550 M/Days @ 500/-	2,75,000/-	1100,000/-	53/592
		Total =	11,00,000/-		
		Supplier by Contractor =	654750/-		
		Total =	1754,750/-	Nil	

Md. Aladdin
Forester

Range Officer Nalchira Range, Hatiya.
Coastal Forest Division, Noakhali..

Md. Faruk Hossain
Forester
Dhalchar Beat
Range-Nalchira, Hatiya.

POST PLANTING MAP OF JAW PLANTATION AREA
 BEAT: DHAL CHAR, RANGE: NALCHIRA
 COASTAL FOREST DIVISION NOAKHALI
 NAME OF CHAR: CHAR AYSA
 YEAR: 2022-23
 AREA: 50.0 HEC.
 SCALE = 1:15840



(Md Farid Meah)
 Deputy Conservator of forests
 Divisional Forest Officer
 Coastal Forest Division, Noakhali.

REF: ~

1	JAW PLANTATION AREA - 2022-23	
2	RIVER	
3	KHAL	
4		

কাজী তারিকুর রহমান
 সহকারী বন সংরক্ষক (নোয়াখালী জেলা)
 উপকূলীয় বন বিভাগ নোয়াখালী

Md. Alauddin
 Range Officer
 Nalchira Range
 Coastal Forest Division, Noakhali

Md Faruk Hossain
 Forester
 Dhalchar Beat
 Range Nalchira, Haty.

মোঃ শাফিকুল ইসলাম
 সার্ভেয়ার
 উপকূলীয় বন বিভাগ নোয়াখালী।

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8.	Name & Designation of the Officers & stuffs Involved in the Plantation Raising Activitiest	1. Mr. Md. Farid Meah, DCF. 2. Mr. Kazi Tariqur Rahman, ACF. 3. Md. Alauddin,Forester, Range Officer 4. Md Faruk Hossain,Forester, Beat Officer. 5. Mr. Samim Al Mamun, Forest Guard. 6. Mr. Babul hoque, Forest Guard. 7. Mr. Jamshed uddin Seddek, Plantation Male.	
B Characteristic of the Plantation Site			
1.	Area of the Plantation	50.0 ha Jhaw Plantation in the year of 2022-2023.	
2	Edaphic Condition	Physiographically the Char is part of the young Meghna Estuarine Flood Plain. Hobgoblin, the site is soft and clayey and it is also comparatively compact and harder as well as at the same time subjected to more saline condition. These soils permit easy penetration of mound and mangrove tree roots because of being quite fertile.	
3	Climate Condition	This char has a tropical monsoon climate, characterized by basically four seasons per year. The average temperature usually ranges between 11 ° to 29 ° C in winter and 21 ° to 38 ° C during summer. The average annual rainfall is 3257 mm (Islam 2009) and varies from 2500 mm to 4500 mm. Cyclones pass regularly, particularly at the beginning and at the end of the monsoon.	

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Sustainable Forests & Livelihoods (SUFAL) Project

“Jhaw Plantation”

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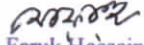
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5	Climate Resilient Aspect of the Plantation	★ The Coastal forest, breakwater, seawall is to absorb the impact forces and to retard the flow of large storm waves and cyclone. A seawall, if tall enough, reflects the waves back out to sea. On the other hand, permeable structures, like breakwaters and coastal forests, partly reflect and partly transmit the water. In the case of a coastal forest, energy is progressively absorbed as it passes through the forest. Without the forest barrier, the tsunami/cyclone will run-up to a maximum height determined by the magnitude and nature of the seismic event that created the tsunami and local factors such as the coastal profile, offshore bathymetry and beach slope that modify the wave's force. Once the tsunami comes on shore, the amount of reduction in water depth, velocity, and force depends on how much water is reflected and energy is absorbed by the coastal forest. ★ Through the afforestation in the project area biodiversity conservation, human security and so on will be ensured in the project area. Therefore, greenbelt of the project area by rising said plantation has been planned to bring back the balance of ecosystem into the same char lands, and to develop a friendly environment by raising coastal afforestation. It is for the fact that the land is suitable for mangrove and non-mangrove plantations where bio-physical and socio-economic conditions will improve. The Jahajmara Sadar will be brought to work as a barrier during natural disasters by dint of the mangrove plantation.
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4	Post-Planting Map (Scale)	Post-Planting Map has been Attached in this Journal

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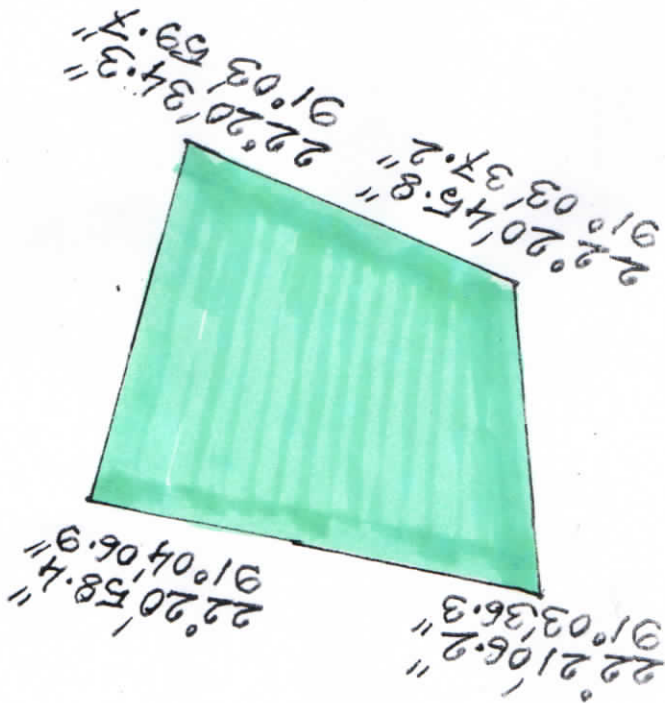
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Md. Alauddin
Forester

Range Officer Nalchira Range, Hatiya.
Coastal Forest Division, Noakhali..

Md. Faruk Hossain
Forester
Dhalchar Beat
Range-Nalchira, Hatiya.

POST PLANTING MAP OF JAW PLANTATION AREA
 BEAT: DHAL CHAR, RANGE: NALCHIRA
 COASTAL FOREST DIVISION NOAKHALI
 NAME OF CHAR: CHAR AYSA
 YEAR: 2022-23
 AREA: 50.0 HEC.
 SCALE = 1:15840



(Md Farid Meah)
 Deputy Conservator of Forests
 Divisional Forest Officer
 Coastal Forest Division, Noakhali.

REF: ~

1	JAW PLANTATION AREA - 2022-23	
2	RIVER	...
3	KHAL	...
4		

ফোর্স্ট ওয়াশিং স্টেশন
 সফটওয়্যার ও মেশিন (আইটিএস) কোম্পানী
 সফটওয়্যার ও মেশিন (আইটিএস) কোম্পানী

Md. Alauddin
 Range Officer
 Nalchira Range
 Coastal Forest Division, Noakhali

Md Faruk Hossain
 Forester
 Dhalchar Beat
 Nalchira Range, Hany.

ফোর্স্ট ওয়াশিং স্টেশন
 সফটওয়্যার ও মেশিন (আইটিএস) কোম্পানী